IST 691: Research Methodology and Scientific Writing Skills

Non-Experimental Research Approaches

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Outline

- Quantitative vs Qualitative Research Approaches
  - Nonexperimental Quantitative Research
  - Qualitative research
    - Grounded theory
    - Phenomenology
    - Field research (Ethnography, Case study, Others

Quantitative and Qualitative Research Approaches

- Classifying Research
  - Experimental vs Descriptive (i.e, Non-experimental) Research
  - Quantitative vs Qualitative Research

Experimental vs Descriptive Research

- Experimental Research
  - Attempts to identify cause and effect
    - Researcher **manipulates** an independent variable to test its effect on a dependent variable, allowing for the establishment of cause-and-effect relationships
- Descriptive Research
  - Researcher observes variable to be able to describe phenomena
    - **No manipulation** of the variables, hence does not prove causation

Quantitative vs Qualitative Research

- Quantitative vs Qualitative Research
  - Quantitative Research Study
    - Research study that collects numerical data
    - Uses numbers and graphs to test hypotheses and provide generalizable results
      - Focus: Measures and tests theories or hypotheses.
      - Data: Numerical, such as statistics or measurements.
      - Purpose: To confirm theories, identify patterns, and generalize results to a larger population.
      - Methods: Surveys, experiments, and numerical observations.
  - Qualitative Research Study
    - Research study that does not collect numerical data
      - Focus: Explores ideas, experiences, and meanings in detail.
      - Data: Non-numerical, such as text, interviews, or observations.
      - Purpose: To understand concepts, motivations, and emotions; often used to explore new ideas or to provide rich context.
      - Methods: Interviews, focus groups, and case studies

Quantitative vs Qualitative Research

- How to choose?
  - Choose qualitative if you need to explore a topic in-depth or understand a specific experience or perspective
  - Choose quantitative if you need to measure something, test a hypothesis, or establish cause-and-effect relationships
  - Consider using mixed methods research if your research question requires both numerical and in-depth data to be fully answered

Quantitative vs Qualitative Research Approaches

- Quantitative
  - Experimental
  - Non-experimental
    - Correlational study
    - Ex Post Facto study
    - Longitudinal study and Cross-Sectional study
    - Meta analysis
    - The survey
    - Etc.
- Qualitative
  - Phenomenology
  - Ethnography
  - Case study

Nonexperimental Quantitative Research

- Correlational study
- Ex Post Facto study
- Longitudinal study and Cross-Sectional study
- Naturalistic observation
- Meta analysis
- Analysis of existing data
- The survey

Correlational Study

- Non-experimental research method in which a researcher measures two variables and understands and assesses the statistical relationship between them with no influence from any extraneous variable
- Aims to understand the strength and direction of the relationship between two or more variables
- Results could reveal
  - Positive correlation
    - An increase in one variable leads to a rise in the other variable
    - A decrease in one variable leads to a reduction in the other variable
    - Example: The amount of money a person has might positively correlate with the number of cars the person owns
  - Negative correlation
    - Opposite of positive correlation
      - An increase in one variable leads to a fall in the other variable
      - A decrease in one variable leads to an increase in the other variable
    - Example:
      - Educated might negatively correlate with the crime rate when an increase in one variable leads to a decrease in another and vice versa
      - Hence, if a country's education level is improved, it can lower crime rates
        - But correlation does not mean causation, i.e., that lack of education leads to crimes
        - It only means that a lack of education and crime is believed to have a common reason – perhaps poverty
  - No correlation
    - A change in one variable may not necessarily see a difference in the other variable

Ex Post Facto Study (or after-the-fact research)

- Also known as Causal-Comparative Research (or Quasi-Experimental):
- Examines the cause-and-effect relationship between variables of interest but these variables are not subject to manipulation but must be chosen after the fact
  - The investigation starts after the fact has occurred without interference from the researcher
  - Often applied as a substitute for true experimental research
- Often used when an independent variable cannot be manipulated
- An example
  - Determining a cause-and-effect relationship between height and IQ scores
  - Researcher cannot physically alter someone's height, therefore specific groups will be put together based on how tall they are

Longitudinal Study and Cross-Sectional Study

- Longitudinal study
  - Developmental field study that repeatedly measures the same characteristics in a single sample at selected time intervals
  - Developmental field study that measures the same characteristics in representative samples at one point in time

Naturalistic Observation

- A descriptive research technique for unobtrusively collecting data on naturally occurring behaviour
  - You record the behaviours of your research subjects in real world settings
  - You avoid interfering with or influencing any variables in a naturalistic observation
  - Examples
    - Usability research
      - Observing users interacting with a product as a way to understand the usability of the product and to some extent the overall user experience
    - Consumer research
      - You study how grocery shoppers navigate a store and shop differently after a layout change
    - Child development
      - You track language development in a child's natural environment (their own home) with an audio recording device

Meta Analysis

- A quantitative technique that is used to integrate and describe a large number of studies
- Why meta analysis
  - To establish statistical significance with studies that have conflicting results
  - To develop a more correct estimate of effect magnitude

Analysis of Existing Data

- Utilizes pre-existing datasets to answer a research question
- Can be more cost-effective and efficient, as the data has already been collected.

Survey Research

- Non-experimental research in which an interview technique is used to gather data on a given state of affairs in a representative sample of the population
- May be quantitative or qualitative
- Two important characteristics
  - The variables of interest are measured using self-reports (using questionnaires or interviews)
  - Considerable attention is paid to the issue of sampling
    - Survey researchers have a strong preference for large random samples because they provide the most accurate estimates of what is true in the population

Survey Research – Data Collection Strategies

- Depends on both the researcher's goals and the budget
- In-person interviews
  - Personal contact can be important, for example, when the interviewer must see and make judgments about respondents, as is the case with some mental health interviews
  - Highest response rates and provide the closest personal contact with respondents
  - However, most costly approach
- By telephone
  - Cost not as high as in-person interviews
  - Lower response rates than interviews, but still provides some personal contact with respondents
  - In the era of land lines, telephone directories provided fairly comprehensive sampling frame
  - Because most people choose to only have cell phones, the telephone directories advantage is no longer there

Survey Research – Data Collection Strategies

- Mail surveys
  - Less costly still but generally have even lower response rates, making them most susceptible to non-response bias
- Over the internet
  - Have become very common
  - Increasingly easy to construct and use
  - Initial contact by mail, with a link provided to the survey
  - Can work well when the population consists of the members of an organization who have known email addresses and regularly use them
  - For other populations, it can be difficult or impossible to find a comprehensive list of email addresses to serve as a sampling frame
  - Low cost

Survey Research – Questionnaire Construction

- Important to construct a number of questions that will provide an answer to the research question
- This is not an easy task, as the answers people give can be influenced in unintended ways by
  - The wording of the items
  - The order of the items
  - The response options provided
  - Many other factors
- These influences may
  - Add noise to the data (best case)
  - Result in systematic biases and misleading results (worst case)

Survey Research – Questionnaire Construction

- Example of difficulty in questionnaire construction
  - Question
    - How many alcoholic drinks do you consume in a typical day?
  - Answer options
    - A lot more than average
    - Somewhat more than average
    - Average
    - Somewhat fewer than average
    - A lot fewer than average
- Difficulties in interpreting the question
  - Do “alcoholic drinks” include beer and wine (as opposed to just hard liquor)
  - What is a “typical day”? A typical weekday, a typical weekend day, or both

Survey Research – Questionnaire Construction

- Difficulties in retrieving relevant information from memory to answer the question
  - Respondent might think vaguely about some recent occasions on which they drank alcohol
    They might carefully try to recall and count the number of alcoholic drinks they consumed last week
  - They might retrieve some existing beliefs that they have about themselves (e.g., “I am not much of a drinker”), then use this information to arrive at a tentative judgment about how many alcoholic drinks they consume in a typical day
    - This mental calculation might mean dividing the number of alcoholic drinks they consumed last week by seven to come up with an average number per day
- Difficulties in providing an answer
  - Respondents must format any tentative answer to the question in terms of the response options actually provided
  - In this example, the options pose additional problems of interpretation
    - What does “average” mean?
    - What would count as “somewhat more” than average?
  - Finally, they must decide whether they want to report the response they have come up with or whether they want to edit it in some way
    - For example, if they believe that they drink a lot more than average, they might not want to report that for fear of looking bad in the eyes of the researcher, so instead, they may opt to select the “somewhat more than average” response option.

Survey Research – Types of Research Questions

- Open-ended
- Close-ended

Open-Ended Questions

- Questions that enable respondents to answer in any way they please
- Relatively easy to write because there are no response options to worry about
- More qualitative in nature
- Examples
  - What is the most important thing to teach children to prepare them for life?
  - Please describe a time when you were discriminated against because of your age.
  - Is there anything else you would like to tell us about?
- Pros
  - Unbiased and do not provide respondents with expectations of what the researcher might be looking for
  - Useful when researchers do not know how participants might respond
- Cons
  - Respondents are more likely to skip open-ended items because they take longer to answer
  - They are more difficult for the researcher to analyze because the answers must be transcribed, coded, and submitted to some form of qualitative analysis, such as content analysis

Close-Ended Questions

- Questions that require respondents to choose from a limited number of predetermined responses
  - How old are you?
    - Under 18
    - 18 to 34
    - 35 to 49
    - 50 to 70
    - Over 70
- Used when researchers have a good idea of the different responses that participants might make
- More quantitative in nature
- More difficult to write because they must include an appropriate set of response options
- However
  - They are relatively quick and easy for participants to complete
  - They are much easier for researchers to analyze because the responses can be easily converted to numbers

Ordering of Questions

- Ask the most important questions first when the respondent is more active
- Ask those questions which are most sensitive to conditioning such as attitudes and preferences earlier
- Ask factual and historical information towards the end as respondent becomes less enthusiastic and fatigued

Questionnaire Length

- Questionnaire length
  - This is the amount of time it takes a respondent to complete a questionnaire
- Questionnaire length is important because it can directly affect response rates, survey costs, and data quality
  - Longer questionnaires
    - Higher data collection costs and greater respondent burden and may lead to lower response rates and diminished quality of response
    - However, below a certain threshold, questionnaire length appears to bear little relationship to response rate or data quality

Obtaining Participants for a Survey

- Population
  - The entire group that you want to draw conclusions about
- Sample
  - The specific group that you will collect data from
    - Sample size is always less than the size of the population
- Several approaches to select the sample
  - Convenience Sampling (also called haphazard sampling)
    - Non-probability sampling method
    - Based on convenience
    - Used when there are only a few available members of the target population who can become the participants in the survey
    - Results not generalizable to the population
  - Random sampling
    - Sampling in which every member of the population has an equal chance of being selected for the study
    - A random sampling technique must be used if we need to generalize results to a population

Qualitative Research

- Investigates people in their natural environments
- Three components
  - Interpretative
    - Qualitative data consists of words, pictures, clothing, documents, etc. that need to be interpreted
  - Multimethod
    - A variety of methods are used to collect data
      - Individual account of personal experience, introspective analysis, interviews with an individual, observation of people, written documents, historical information, photographs, etc.
  - Conducted in the field, or a person's natural surroundings

Qualitative Research

- In qualitative research
  - We do not test hypothesis or previous theories
  - We may try to develop new theories based on what happens in specific situations
  - We do not try to generalize our findings
  - We often rely on data collected from interviews, observations, and content analysis of newspapers, books, videos, case records, and other already developed documents
  - We usually do not know or try to develop response categories prior to conducting the study

Common Types of Qualitative Research

- Grounded theory
- Phenomenology
- Ethnography
- Case study

Grounded Theory (GT)

- Research study that is concerned with the generation of theory from the data itself
  - GT is used to uncover such things as social relationships and behaviours of groups, known as social processes
  - Researcher collects and analyzes data simultaneously, identifying patterns and developing a theory that is "grounded" in the participants' experiences.
- Such a study
  - Likely to begin with a question, or even just with the collection of qualitative data
  - Reviewing the data leads to ideas or concepts which "emerge" from the data
  - These ideas/concepts are tagged with codes that succinctly summarize the ideas/concepts
  - With more data collected and re-reviewed, categories emerge
  - These categories become the basis of a hypothesis or a new theory
    - Contrast with traditional scientific model of research, where the researcher chooses an existing theoretical framework, develops one or more hypotheses derived from that framework, and only then collects data for the purpose of assessing the validity of the hypotheses

Phenomenology

- Phenomenology
  - Philosophy of experience
  - Explores the lived experiences of individuals to understand the meaning and essence of a particular phenomenon
  - Hence this approach to research involves
    - Description of an individual, or group of individuals' conscious experience of a phenomenon
    - Could involve interviewing people who have experienced a specific life event, such as a major illness, to understand their perceptions and feeling

Ethnography

- Ethnography
  - The scientific description of peoples and cultures with their customs, habits, and mutual differences.
  - Involves immersive fieldwork and participant observation to understand the culture of a group
    - Example, a researcher might live within a community for an extended period to observe and document its social structures, beliefs, and behaviors
  - Data are collected through observations and interviews, which are then used to draw conclusions about how societies and individuals function

Case Study

- An in-depth investigation of a single individual, group, event, or community
  - Example, an in-depth study of a small business's challenges during the pandemic, gathering data from interviews, internal documents, and observations

Strengths and Weaknesses of Qualitative Research

- Strengths
  - Depth of understanding
  - Flexibility
- Weaknesses
  - Subjectivity
  - Suggestive, not definitive
  - Limited generalizability
- Mixed methodology is possible